Ex) Let U be a Universal set and $n(U) = 12$. If $A, B \subseteq U$ are such that $n(B) = 6$ and $n(A \cap B) = 2$ then $n(A \cup B')$ is equal to?

→ Given,

$n(U) = 12$

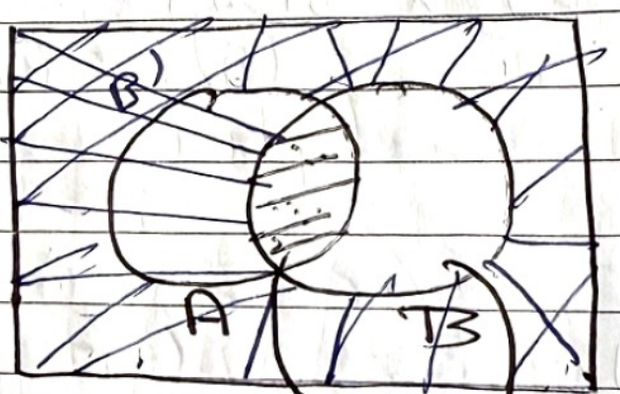
$A, B \subseteq U$

$n(B) = 6$

$n(A \cap B) = 2$

$n(A \cup B') = ?$

→



2 → $(A \cap B)$

4 → 6

$n(A \cup B') = 8$

Ex) Find $A-(A-B)$ if $A = \{5, 9, 13, 17, 21\}$ and $B = \{3, 6, 9, 12, 15, 18, 21, 24\}$

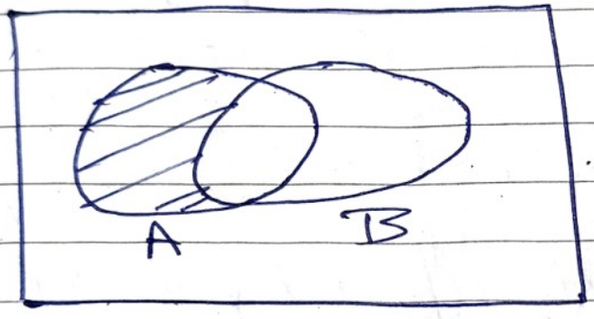
Given.

→ $A-(A-B) \Rightarrow ?$

$A \Rightarrow \{5, 9, 13, 17, 21\}$

$B \Rightarrow \{3, 6, 9, 12, 15, 18, 21, 24\}$

1) $A-B \Rightarrow$ ~~{9, 21}~~ $\{5, 13, 17\}$



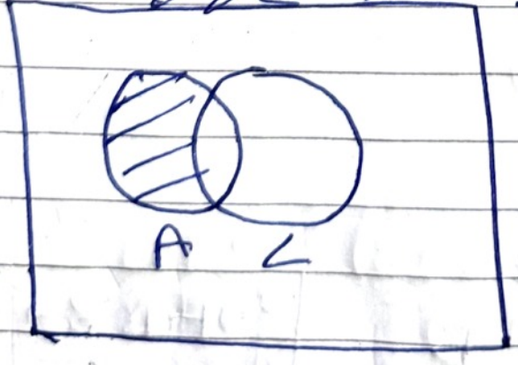
2) $A-(A-B)$

↳ assume it in a new set and we name it C

$A - C$

$C \Rightarrow$ ~~{9, 21}~~ $\{5, 13, 17\}$

$A \Rightarrow$ ~~{5, 9,~~ 13, 17, 21}



$A - C \Rightarrow$ ~~{5, 13, 17}~~ $\{9, 21\}$

Ex. Let P(x) denote the Power Set of a Set X. for any two Set A and B if $A \neq B$, $P(A) \neq P(B)$, then.

A) $A \cup B \rightarrow A \Delta B$
B) $A = B$
C) $A \cap B = \phi$
D) $A \Delta B = \phi$

* Two Power Set can be Equal.

If,

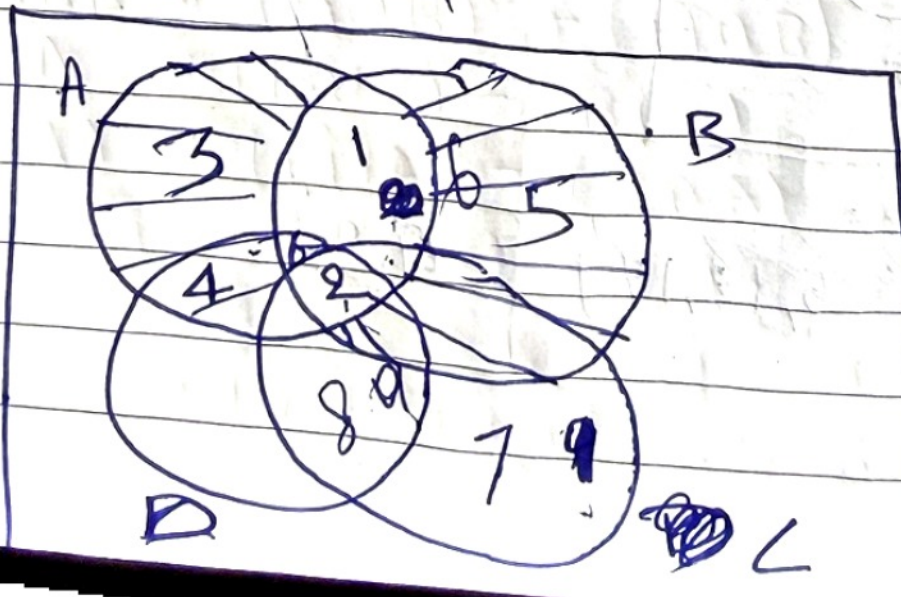
$P(A) = P(B)$
then and only then
$A = B$

Ex If $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 5, 6\}$, $C = \{2, 7, 8, 9\}$ and $D = \{2, 4, 8, 8\}$

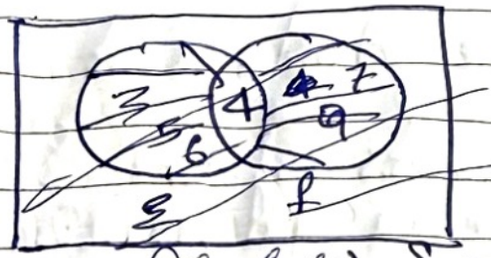
then $(A \Delta B) \Delta (C \Delta D) = ?$

Given,

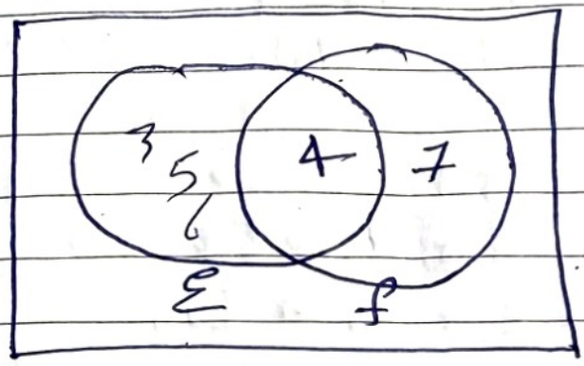
→ $A = \{1, 2, 3, 4\}$
$B = \{1, 2, 5, 6\}$
$C = \{2, 7, 8, 9\}$
$D = \{2, 4, 8, 8\}$



$\xi \Rightarrow A \Delta B = \{3, 4, 5, 6\}$
$f = C \Delta D = \{4, 7, 9\}$



$E \Delta f \Rightarrow \{3, 5, 7, 9\}$



$E \Delta f \Rightarrow \{3, 5, 6, 7\}$

Ex) Let $S \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $A \Rightarrow \{2, 3, 4, 5, 6\}$. Then the no. of subsets B of S such that $A \Delta B \Rightarrow \{5\}$ is

⇒ Given,

$S \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$A \Rightarrow \{2, 3, 4, 5, 6\}$

$B \Rightarrow$ What kind of subset B should become where the ans should be only $\{5\} \Rightarrow A \Delta B$

1

→ $\{5\}$ Subset is only possible

Eg. A, B, C are three finite Set Such that $A \cap B \cap C$ has 10 elements. if the Sets $A \Delta B$, $B \Delta C$ and $C \Delta A$ have 100, 150 & 200 elements, respectively, then the number of elements in $A \cup B \cup C$ is.

Given,

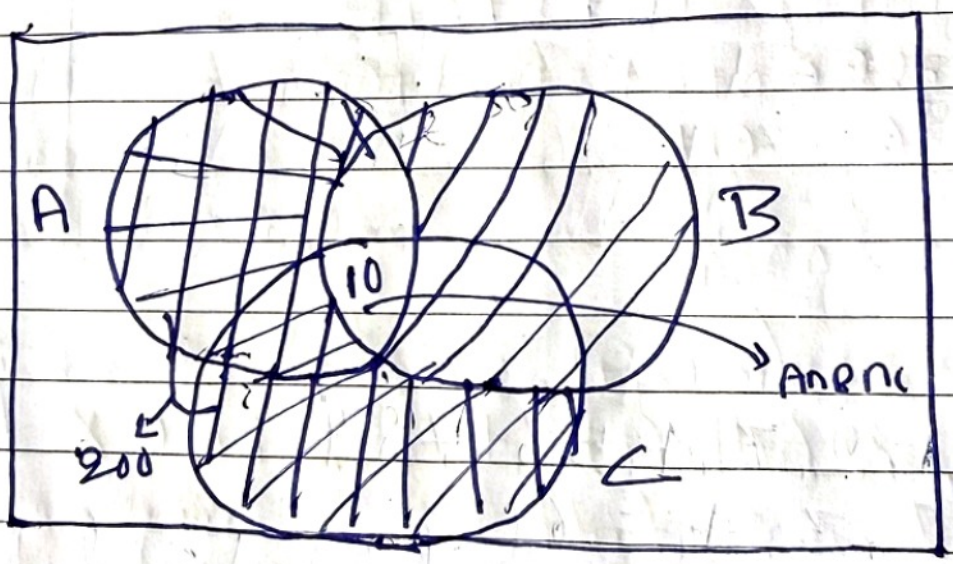
→ $A \cap B \cap C \Rightarrow 10$

$A \Delta B \Rightarrow 100$

$B \Delta C \Rightarrow 150$

$C \Delta A \Rightarrow 200$

1)



→ $A \cap B \cap C \Rightarrow 10$

$A \cap (B \cap C) \Rightarrow 10$ — (1)

$A \Delta B \Rightarrow (A \cup B) - (A \cap B) \Rightarrow 100$ — 2

$B \Delta C \Rightarrow (B \cup C) - (B \cap C) \Rightarrow 150$ — 3

$C \Delta A \Rightarrow (C \cup A) - (C \cap A) \Rightarrow 200$ — 4

$(B \cup C) - (B \cap C) \Rightarrow 150$

Can be written as

$-B \cap C \Rightarrow 150 - (B \cup C)$

(-) on Both sides

$(B \cap C) \Rightarrow (B \cup C - 150)$ ——— (5)

Replace "5" in (1)

$A \cap (B \cup C - 150) \Rightarrow 10$

$A \cap B \cup C - 150 \Rightarrow 10$

$A \cap B \cup C - 150 \Rightarrow 10$

$(A \cap B) \cup C \Rightarrow 160$ ——— (6)

To find $(A \cap B)$ what we can do is Simplify Eq (2) where

$(A \cup B) - (A \cap B) \Rightarrow 100$

$-(A \cap B) \Rightarrow 100 - A \cup B$

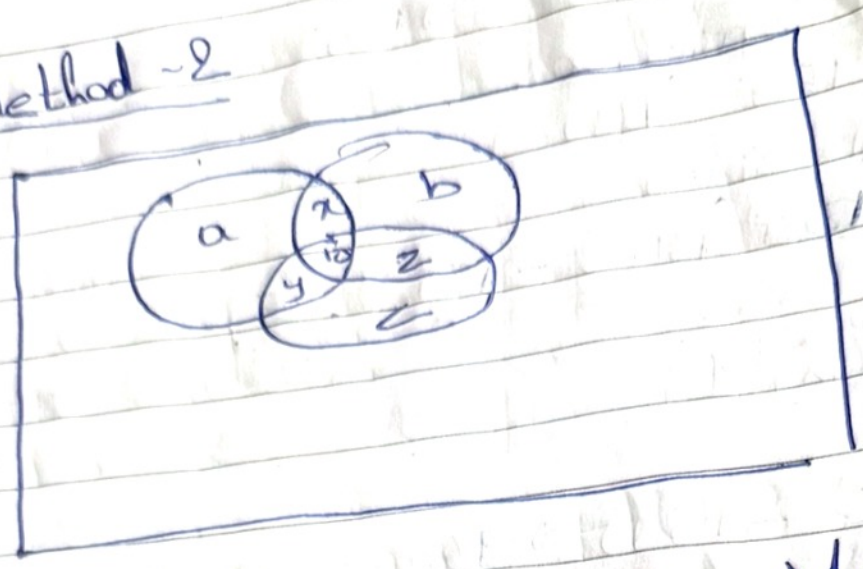
$(A \cap B) \Rightarrow A \cup B - 100$ ——— (7)

Replace (7) in (6)

$((A \cup B) - 100) \cup C \Rightarrow 160$

$A \cup B \cup C \Rightarrow 260$

Method -2



$n(A \Delta B) = a+y+b+z \Rightarrow 100$

$n(B \Delta C) \Rightarrow x+b+c+z \Rightarrow 150$

$n(C \Delta A) \Rightarrow a+x+c+y \Rightarrow 200$

$+2a+2x+2c+2y+2z+2b \Rightarrow 450$

$2(a+x+y+z+c+b) \Rightarrow 450$

$a+x+y+z+c+b \Rightarrow 225 + 10$

$+10 \Rightarrow 235$

ξ

$A \cap B \cap C$

Ex Let $X \Rightarrow \{1,2,3,4,5\}$. The no. of different ordered pair (Y, Z) that can be formed such that $Y \subseteq X$, $Z \subseteq X$ and $Y \cap Z$ in ϕ is

→ Given,

$X \Rightarrow \{1,2,3,4,5\}$

Subset of $X \Rightarrow$

$2^n \Rightarrow 2^5 \Rightarrow 32$

$\Rightarrow \{\}; \{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{1,2\}, \{1,3\}, \{1,4\}, \{1,5\}, \{2,3\}, \{2,4\}, \{2,5\}, \{3,4\}, \{3,5\}, \{4,5\}, \{1,2,3\}, \{1,2,4\}, \{1,2,3\}, \{1,5,4\}, \{1,3,5\}, \{2,3,4\}, \{2,3,5\},$

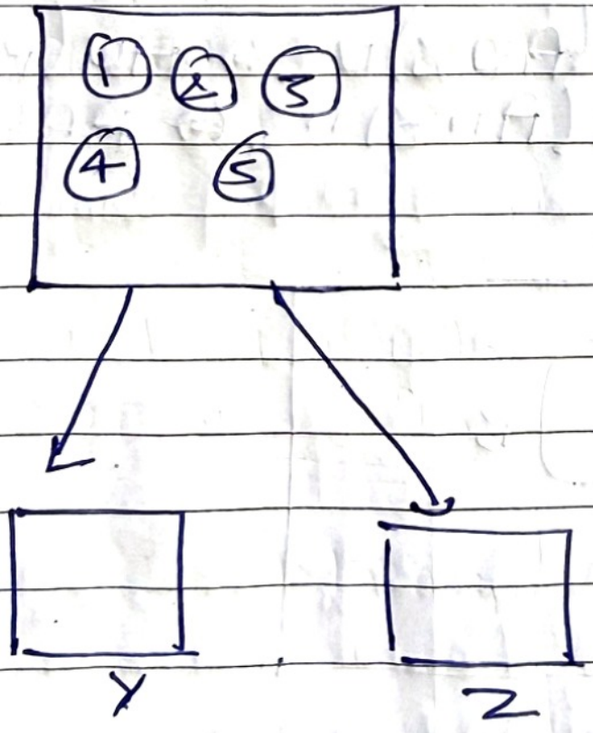
{3, 4, 5} ___ 8 more}

2 order pair We have.

→ {1, 2} (=) {2, 1}
{1, 3} (=) {3, 1}
{1, 4} (=) {4, 1}
{1, 5} (=) {5, 1}
{2, 3} (=) {3, 2}
{2, 4} (=) {2, 4}
{2, 5} (=) {2, 5}
{3, 4} (=) {3, 4}
{3, 5} (=) {3, 5}
{4, 5} (=) {4, 5}

10 + 3
⇒ 13

Let play the game of choices.



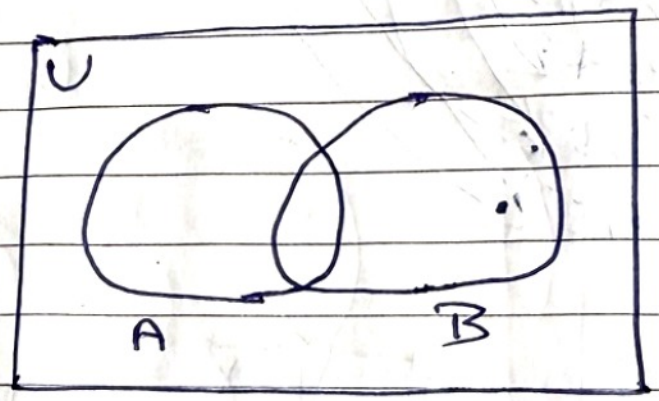
{1, 2, 3, 4, 5} all have "3" choices either they help Y (or) Z (or) they don't help both of them to create an subset so each of them have 3 choices now we multiply that together We get

3 x 3 x 3 x 3 x 3 ⇒ 3^5

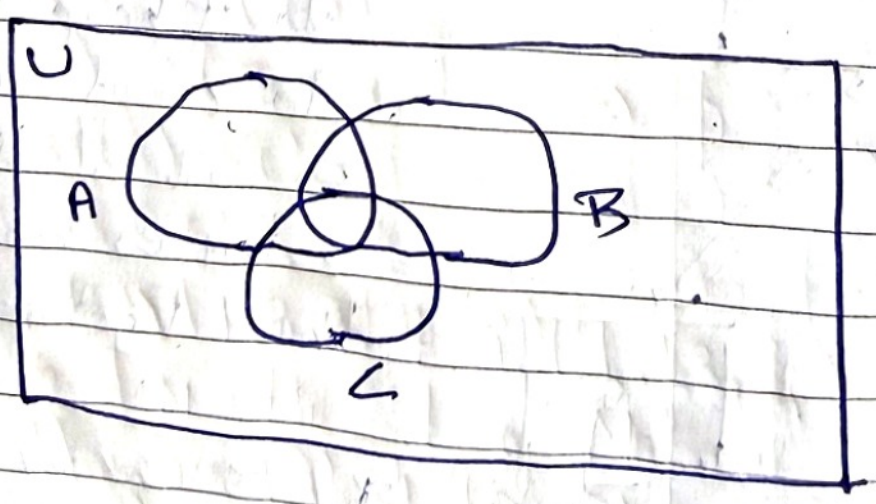
Laws of Algebra of Sets

Properties of Sets

1) Commutative Law: $A \cup B \Leftrightarrow B \cup A$;
$A \cap B \Leftrightarrow B \cap A$

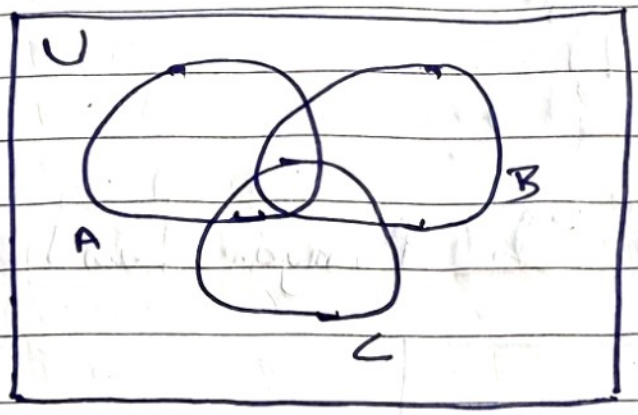


2) Associative Law: $(A \cup B) \cup C \Leftrightarrow A \cup (B \cup C)$
$(A \cap B) \cap C \Leftrightarrow A \cap (B \cap C)$



3) Distributive Law ⇒ $A \cup (B \cap C) \Leftrightarrow (A \cup B) \cap (A \cup C)$
$A \cap (B \cup C) \Leftrightarrow (A \cap B) \cup (B \cap C)$

{ Think of
$A \times (B + C) \Rightarrow (A \times B + A \times C)$ }



4) De morgan's Law ⇒ $(A \cup B)' \Rightarrow (A' \cap B')$
$(A \cap B)' \Rightarrow (A' \cup B')$

{ Think
'whenever there is a ~~Complement~~ive Complement as a power whatwill happen to the inner values is that they will get the Complement as power when you open them and the sign will change as well

5) Identity Law: $A \cup \phi \Rightarrow A$
$A \cap U \Rightarrow A$;

6) Complement Law; $A \cup A' \Rightarrow U$,
$A \cap A' \Rightarrow \phi$
$(A')' \Rightarrow A$

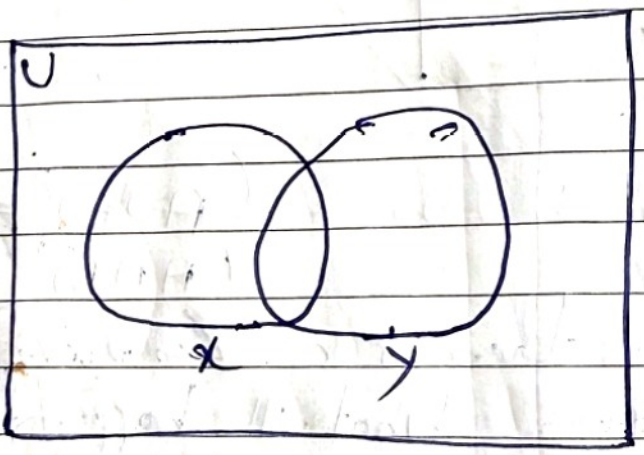
7) Idempoten Law: $A \cap A \Rightarrow A$,
$A \cup A \Rightarrow A$

Ex) If X and Y are two Sets and X' denotes the Complement of X, then $X \cap (X \cup Y)'$ equals.

→ Given,
X' is a Complement of X

find → $X \cap (X \cup Y)'$

$\Rightarrow (X \cup Y)' \Rightarrow (X' \cap Y')$ { De Morgan's law }



→ $X \cap (X' \cap Y') \Rightarrow (X \cap X') \cap Y'$ → ϕ ⇒ from complement law

$\Rightarrow Y' \cap \phi$ { Complement Law }

$\Rightarrow \phi$

Ex If $A \Rightarrow \{1, 3, 5, 7, 9, 11, 13, 15, 17\}$, $B \Rightarrow \{2, 4, \ldots, 18\}$ and N the Set of Natural numbers is the universal Set, then $A' \cup ((A \cup B) \cap B')'$ is

⇒ Given,

$A' \cup ((A \cup B) \cap B')$

$A \Rightarrow \{1, 3, 5, 7, 9, 11, 13, 15, 17\}$

$B \Rightarrow \{2, 4, \ldots, 18\}$

N is a Natural Number as a Universal Set.

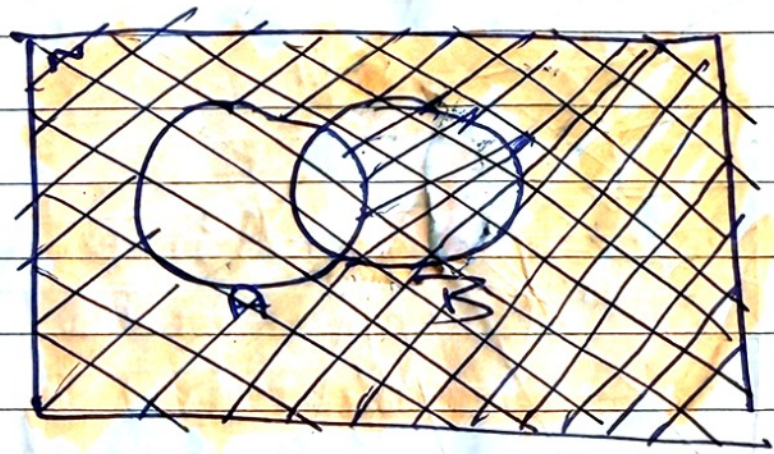
→ $A' \cup (B' \cap (A \cup B))$

→ $A' \cup ((B' \cap A) \cup (B' \cap B))$

$B' \cap B \Rightarrow \phi$

→ $A' \cup ((B' \cap A) \cup \phi)$

→ ~~$(A' \cup (B' \cap A))$~~



→ $A' \cup (\phi \cup (B' \cap A))$

$A' \cup ((\phi \cup B') \cap (\phi \cup A))$

$A' \cup ((B') \cap (A))$

$A' \cup (B' \cap A)$

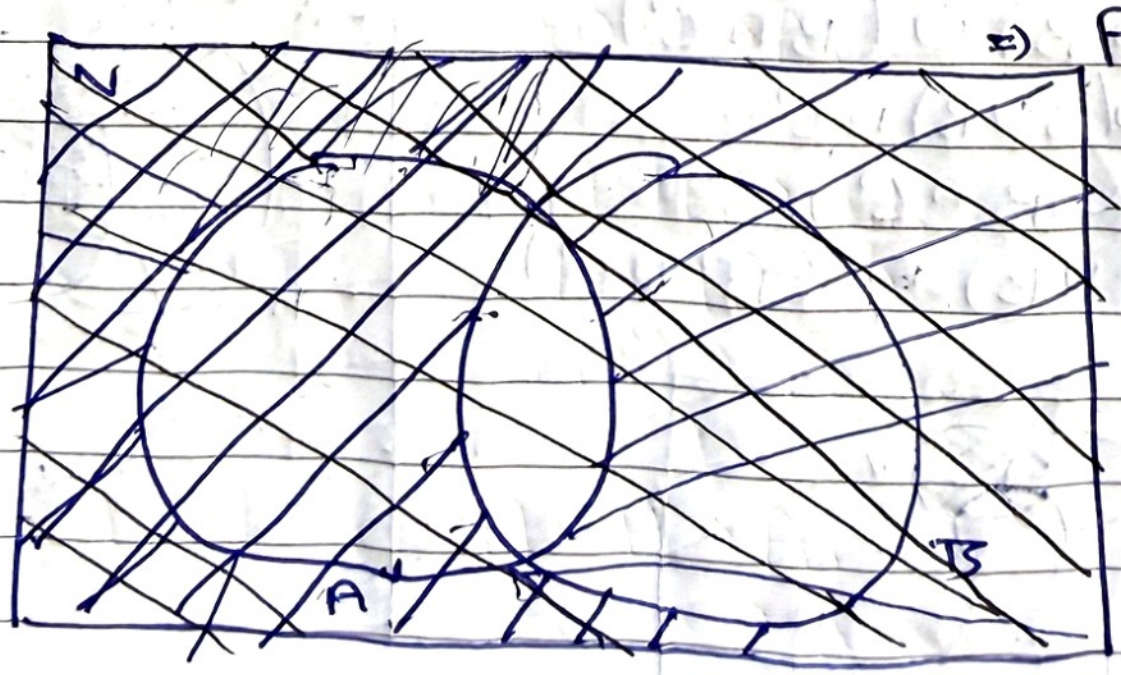
$(A' \cup B') \cap (A' \cup A)$

$(A' \cup B') \cap N \Rightarrow N \cap (A' \cup B') \Rightarrow (N \cap A') \cup (N \cap B')$

$\Rightarrow A' \cup B'$

$\Rightarrow (A \cap B)'$

$\Rightarrow (\phi)' \Rightarrow N$

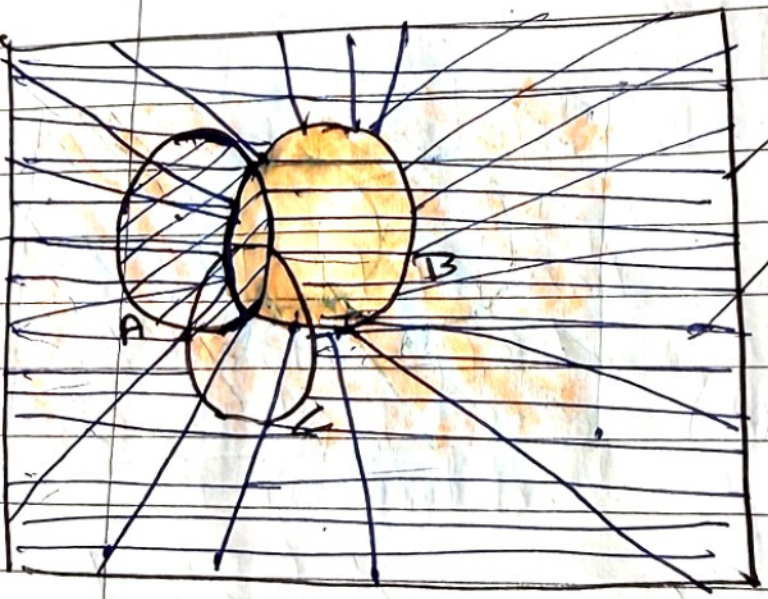


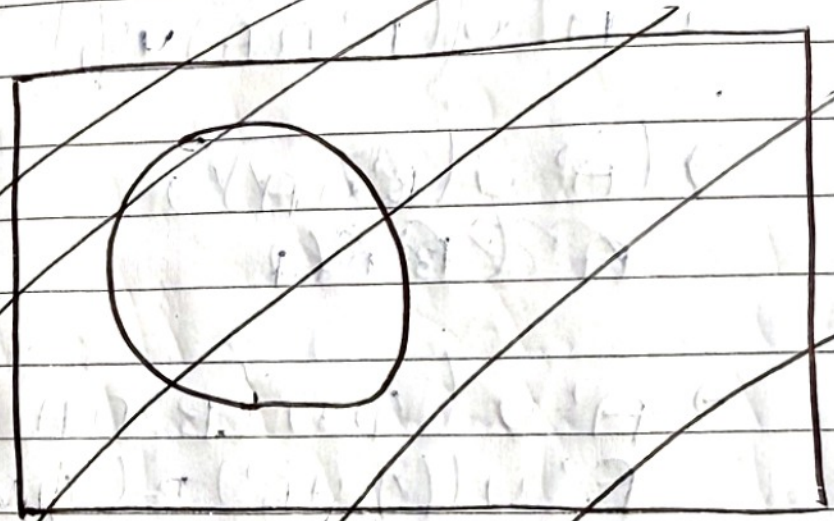
Ex) The Set $(A \cap B')' \cup (B \cap C')$ is Equal to

⇒ Given

$(A \cap B')' \cup (B \cap C') \Rightarrow ?$

→ $(A \cap B')' \cup (B \cap C')$





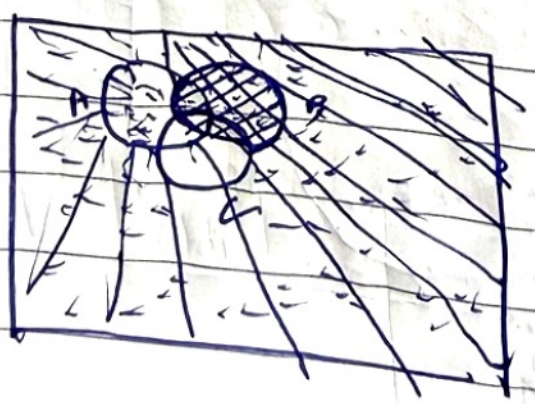
→ $(A \cap B')' \cup (B \cap C')$

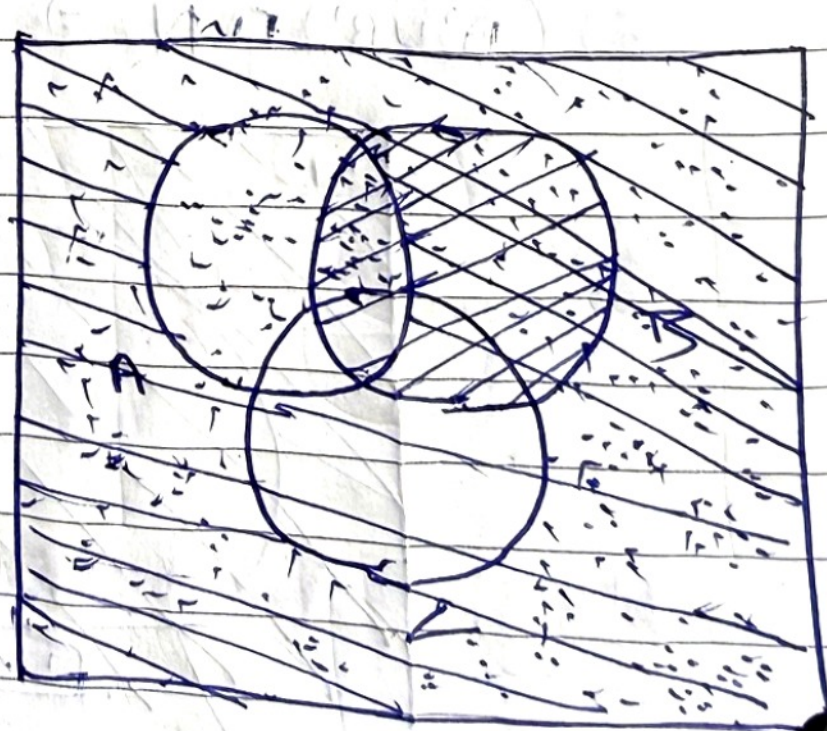
$(A' \cup B) \cup (B \cap C')$

$(A') \cup (B \cup B) \cap C'$

$A' \cup (B \cap C')$

$(A' \cup B) \cap (A' \cup C')$





⇒ $A' \cup$

Ex. The Set $(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$ is equal to.

→

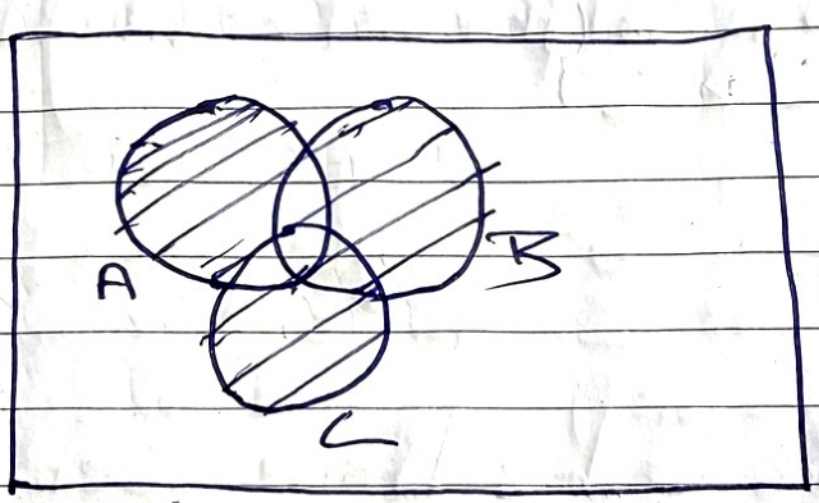
Given,

$(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$

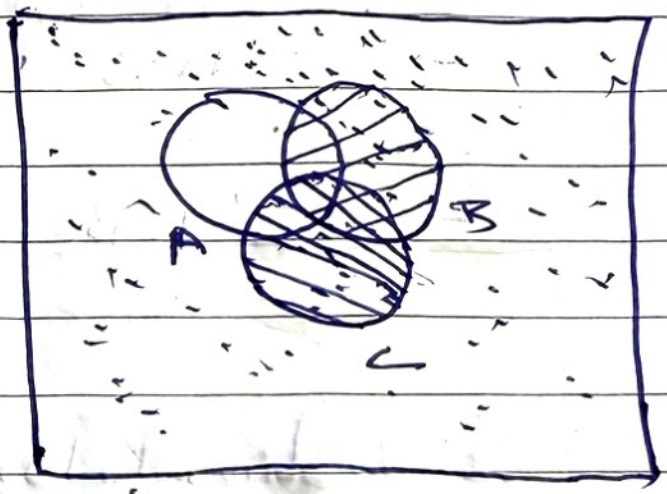
↓

$(A' \cup B \cup C) \cap C'$

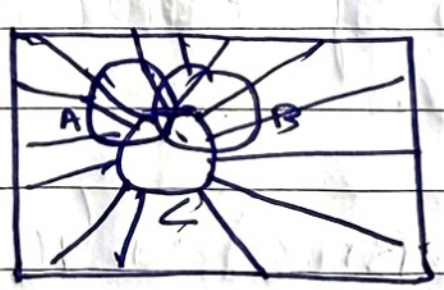
⇒

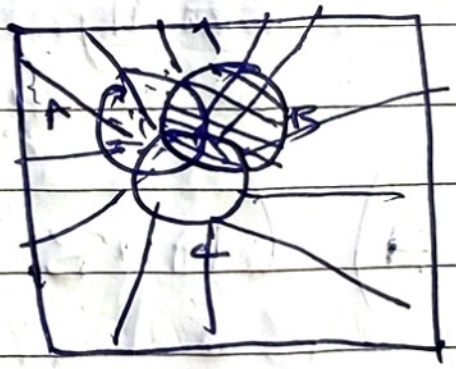


$(A \cup B \cup C)$



$(A' \cup B \cup C)$





⇒ $C' \cap (A' \cup B \cup C) \cap (A \cup B \cup C)$

$((C' \cap A) \cup (C' \cap B) \cup (C' \cap C)) \cap (A \cup B \cup C)$

$\Rightarrow (\phi \cup (C' \cap B) \cup (C' \cap A)) \cap (A \cup B \cup C)$

$\Rightarrow ((\phi \cup C') \cap (\phi \cup B) \cup (C' \cap A)) \cap (A \cup B \cup C)$

$\Rightarrow (C' \cap B) \cup (C' \cap A)) \cap (A \cup B \cup C)$

$\Rightarrow (B \cup A) \cap$

→ $(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$

→ $(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$

→ $(A \cup B \cup C) \cap (A' \cup B \cup C) \cap C'$

→ $(A \cap A') \cup (B \cup C) \cap C'$

ϕ

→ $\phi \cup (B \cup C) \cap C'$

→ $(B \cup C) \cap (C')$

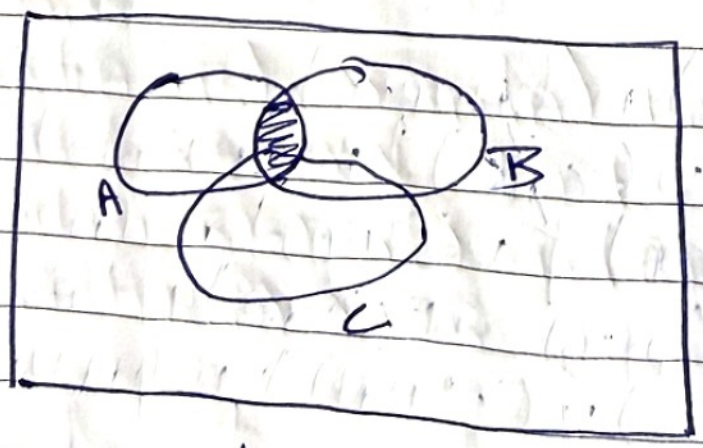
→ $(C' \cap B) \cup (C' \cap C)$ ϕ

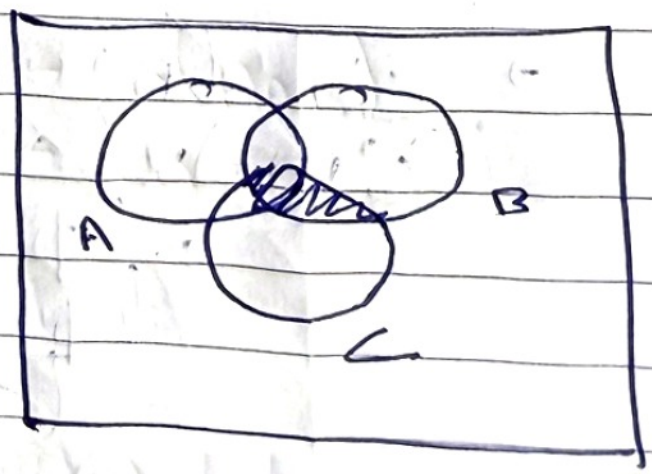
→ $C' \cap B$

Ex Let A, B, and C be three Sets and X be the Set of all elements which belongs to Exactly two of the Sets A, B and C, then X is Equal to.

A) $(A \cap B) \cup (B \cap C) \cup (C \cap A)$

i)

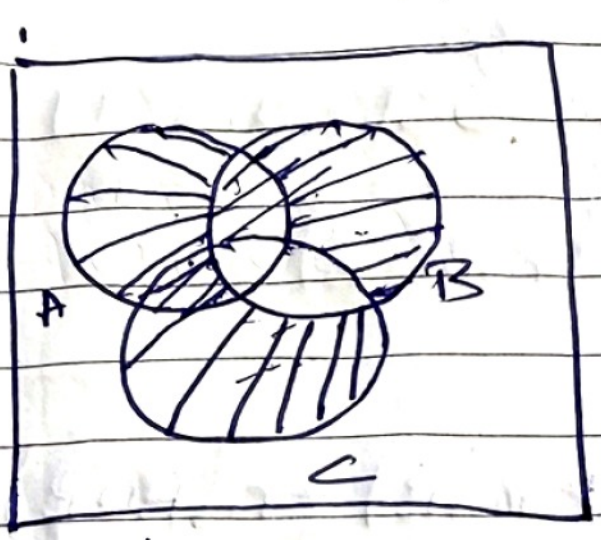




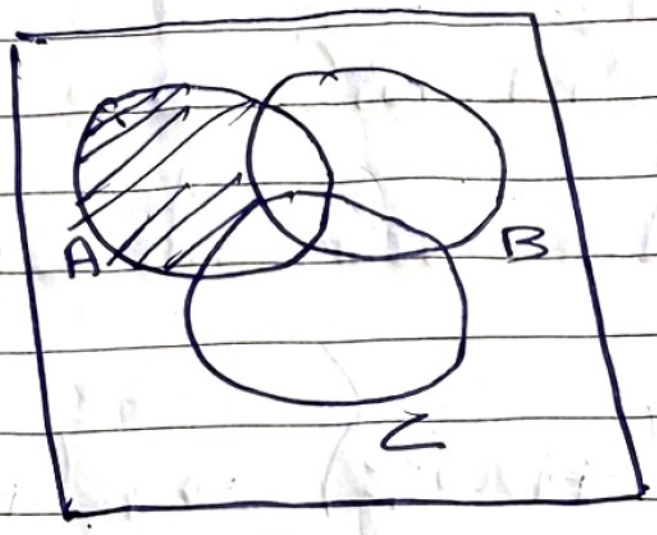
ii) not possible

B) $A \Delta (B \Delta C)$

=>



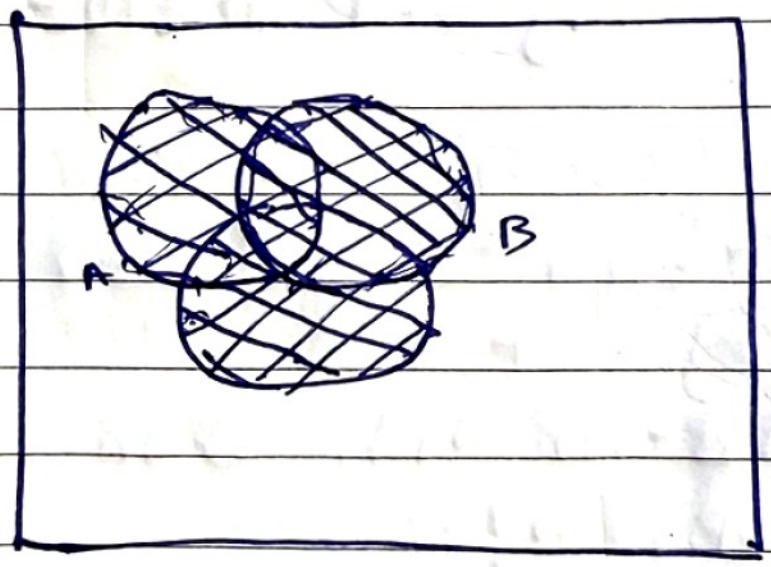
$(B \Delta C)$



A

C) $(\underline{A \cup B}) \cap (\underline{B \cup C}) \cap (C \cup A)$

=>



D) $(A \cup B \cup C) - ([A \Delta (B \Delta C)])$

